Group 1 Activity Packet — RSA Encryption

"SecureBank Transfer" — self-run, Judge-led, no teacher intervention needed

How This Document Works

This packet is designed so your group can run the entire activity yourselves, without waiting for the teacher at any point. Follow the stages below IN ORDER — each stage tells you exactly what you receive, what to do with it, and who to hand your result to next.

- The teacher will not check your work step-by-step during the activity. You are expected to self-check as you go (see the yellow boxes) and cross-check with your teammate(s) (see the blue boxes).
- Where you see a blank line, that is YOUR job to work out — the answer is deliberately not printed anywhere in this document. If you get stuck, re-read the relevant section of your Study Material.
- The Judge / Timekeeper (the last stage) is in charge of running your group's clock, confirming the final results match, scoring part of the rubric themselves using the guidance printed in their own stage, and filling in the Result Card at the end — that Result Card is what gets submitted for grading, and the teacher's script totals it automatically.

Do not skip ahead: Only read the stage that applies to your own role. Reading a later stage's instructions before it's your turn defeats the purpose of the activity — you'll be found out anyway, because the numbers won't make sense to you without the steps in between.

The Scenario

SecureBank's finance office must send a one-digit transaction authorisation code to the Clearing House over a public channel. RSA public-key encryption is used so no shared secret ever has to travel over that channel. A rival firm's Cyber Intelligence Unit intercepts everything sent over the channel — the ciphertext and the public key — and must independently work out the private key well enough to decrypt the message themselves.

Roles in Your Group (10 students)

- Student 1 — Key Generator 30
- Students 2, 3 — Sender A & B 10, 29
- Student 4 — Courier / Channel 6
- Students 5, 6 — Receiver A & B 8, 26
- Students 7, 8, 9 — Attacker A, B, C 1, 2, 18
- Student 10 — Judge / Timekeeper 22

Given Data (the only numbers you start with)

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Secret code to send: M = 4
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Every other number in this activity (the primes, the keys, the ciphertext) must be worked out by the roles below — none of it is given upfront.

Stage-by-Stage Instructions

Stage 1 — Key Generator (Student 1) Varun Vinod .

You start with: Nothing except this instruction sheet — you start the activity.

What to do:

1. Pick two small prime numbers p and q (if unsure, use p=5 and q=11).
2. Compute $n = p \times q$.
3. Compute $\varphi(n) = (p-1)(q-1)$.

4. Pick a small e that is coprime with φ(n) (has no common factor with it other than 1) — e=3 works for the suggested p,q.
5. Find d such that (e × d) mod φ(n) = 1. You can find this by trying d = 1, 2, 3, ... and checking each one.

Self-check before you hand off: Before announcing anything, test your own keys: pick any small test digit (not the real secret code), encrypt it using your e and n, then decrypt your own result using your d and n. If you get the same digit back, your keys are correct — announce them. If you don't, you've made an arithmetic slip somewhere above; find it before continuing.

Your computed n, φ(n), e, d: 55, 40, 3, 27

Hand off: Announce (e, n) out loud / on the board for everyone including the Attacker team to see. Write (d, n) on a separate slip and hand it ONLY to the Receiver team, unseen by anyone else to the whole group (public) and Receiver team (private). Do not show them this stage's instructions — only give them the value(s) named above.

Stage 2 — Sender (Students 2 & 3) DION BOBY, VARUN P DAS

You receive: The public key (e, n), announced publicly. You already know the secret code M = 4 from the Given Data. — handed to you by Key Generator.

What to do:

6. Compute the ciphertext: C = M^e mod n.
7. Write only the resulting number C on a card — do not write M, e, n, or your working on the card.

Cross-check with your partner(s): Both of you compute C independently, then compare. If your answers don't match, redo the calculation together before handing the card over.

Your computed ciphertext C: 9

Hand off: Hand the card with only C written on it to the Courier. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 3 — Courier / Channel (Student 4) ANUSREE ANILKUMAR

You receive: The ciphertext card. — handed to you by Sender.

What to do:

8. Carry the card from the Sender's table to the Receiver's table.
9. Walk close enough to the Attacker team's table that they have a fair chance to copy the ciphertext down — this is intentional, it represents a public channel.

Hand off: Deliver the card to the Receiver team (and Attacker team copies it as you pass). Do not show them this stage's instructions — only give them the value(s) named above.

Stage 4 — Receiver — legitimate path (Students 5 & 6) ROSH, BIBIN

You receive: Ciphertext C, and the private key (d, n) sealed to you earlier by the Key Generator. — handed to you by Courier.

What to do:

10. Compute M = C^d mod n. Since d is large, use the square-and-multiply method from your Study Material — do not try to compute C^d directly.
11. Record how long this step took you, in minutes, starting from when the Courier handed you the card.

Cross-check with your partner(s): Both of you compute M independently using the same square-and-multiply steps, then compare before reporting.

Your recovered code M and time taken (min): 4 , 18 (min)

Hand off: Report your recovered code and your time to the Judge. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 5 — Attacker — attack path (Students 7, 8 & 9) ADARSH LUCUS, ALBIN JOSEPH, JOSHUA GEO

You receive: Ciphertext C and the public key (e, n) only. You do NOT have d — you must derive it yourself. — handed to you by what you copied off the Courier's card.

What to do:

12. Student 7: factor n into its two prime factors p and q, by trial division (try dividing n by 2, 3, 5, 7, 11, ... until it divides evenly).
13. Student 8: using the p and q found, compute $\varphi(n) = (p-1)(q-1)$, then solve for d such that $(e \times d) \bmod \varphi(n) = 1$, the same way the Key Generator did.
14. Student 9: using the d found, compute M = C^d mod n with square-and-multiply, exactly as the Receiver team did.
15. Record your total time from when you first saw the ciphertext to when you had the recovered code — including the factoring step.

Cross-check with your partner(s): Cross-check each other's step before moving to the next — Student 8 should confirm Student 7's factors multiply back to n before using them; Student 9 should confirm Student 8's d satisfies (e×d) mod φ(n) = 1 before decrypting.

Your recovered code M, total time (min), and factors p, q found:

4 , 18 min 30 sec , p = 5 , q = 11

Hand off: Report your recovered code and your total time to the Judge. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 6 — Judge / Timekeeper — you lead this group (Student 10) MAHADEV SUDESH KUMAR

You receive: Both teams' recovered codes and their times. — handed to you by Receiver team and Attacker team.

What to do:

16. You are in charge of running this activity for your group — keep time for each stage above and keep everyone moving, without waiting for the teacher.
17. Confirm the Receiver's M and the Attacker's M are the SAME value. If they differ, send both teams back to recheck their working — do not just average or guess which is right.
18. Calculate the time gap between the legitimate path and the attack path.

Score before you fill in the Result Card: As Judge, you also score three rows of the rubric yourself, right now, using the guidance below — there is no separate teacher check for these.

19. **Attacker/Claimant correctness (0-5):** full 5 if they independently reached the same correct answer as the legitimate side; partial 2-3 if their method was right but they ran out of time or made an arithmetic slip; 0 if the method itself was wrong.
20. **Attacker/Claimant efficiency (0-3):** reward a systematic method over guessing — e.g. Group 1 trying divisors in order rather than randomly.
21. **Explanation quality (0-3):** before scoring this, pick ONE member at random from each sub-team (not the strongest student) and ask them to explain one step out loud. Full marks only if they can actually explain it, not just read out a number.

Once scored, fill in the Result Card on the next page completely and legibly.

Your Role Recap

Result Card — fill this in and submit

The Judge / Timekeeper fills in this card once both the legitimate path and the attack/claim path are complete and confirmed to match. This card is what your teacher will use to grade your group automatically — fill in every field exactly as instructed, using digits/letters only where shown.

GROUP: 1

STUDENT NAMES / ROLL NUMBERS (all 10, by role):

LEGIT_RESULT (Receiver's recovered code M): 4

LEGIT_TIME_MIN (Receiver's time, in minutes): 18 min

ATTACK_RESULT (Attacker's recovered code M): 4

ATTACK_TIME_MIN (Attacker's total time, in minutes): 18min 30sec

ATTACK_METHOD_NOTE (1-2 lines: how did the Attacker factor n and find d?):
Factored n by trial division (found on 3rd Trial) • Found d using square and multipy method.

ATTACK_RESULT_SCORE (0-5, using the correctness guidance above):
5

ATTACK_EFFICIENCY_SCORE (0-3, using the efficiency guidance above):
3

EXPLANATION_SCORE (0-3, from the random member you asked to explain a step):
3

JUDGE_SUMMARY (1 line: did both match? what was the time gap?):
Both Results matched • The time gap was about 8 sec.

How to submit: Hand this page (or a clear photo of it) to your teacher. If asked to submit digitally, type these exact values into a plain text file named Group1_Result.txt, one "FIELD: value" line per field exactly as labelled above, and send it as your teacher instructs.

Overall Time : 38 min 17 sec.

Student Name / ROLL NUMBER.

Varun Vinod	30	key Generator.
Dion Boby	10	Sender A
Varun P Das	29	Sender B
Anusree Anilkumar	6	Courier / channel
Rosh k Lijo	26	Reciever A
Bibin Alex Tom	8	Reciever B.
~~Attu~~ Adarsh Lucas	2	Attacker A
Aalbin Joseph	1	Attacker B
Joshua Creo Johnson	18	Attacker C
Mahadev Sudesh kumar	22	Judge / Timekeeper.

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very field

Group 1 Judge / Timekeeper — Result Card Guide

"SecureBank Transfer" — for the Judge/Timekeeper (Student 10) only

Your Role Recap

- Keep time for every stage and keep the group moving — you run this activity, not the teacher.
- Confirm the Receiver's recovered code and the Attacker's recovered code are the SAME value before scoring anything. If they differ, send both teams back to recheck their working.
- Score the 3 judgement rows below, then fill in every Result Card field.

Field-by-Field: What To Write On The Result Card

GROUP

Your group number.

```
Example only — NOT your group's real answer:
GROUP: 1
```

STUDENT NAMES / ROLL NUMBERS

List all 10 students by role, so the teacher can match scores to individuals.

```
Example only — NOT your group's real answer:
STUDENT NAMES / ROLL NUMBERS: Key Gen-Roll 4; Sender A/B-Roll 11,19; Courier-Roll 2;
Receiver A/B-Roll 7,15; Attacker A/B/C-Roll 3,9,22; Judge-Roll 14
```

LEGIT_RESULT

The single code the Receiver team reported to you — digit only.

```
Example only — NOT your group's real answer:
LEGIT_RESULT: 7
```

LEGIT_TIME_MIN

How many minutes it took the Receiver team from getting the ciphertext card to reporting their result.

```
Example only — NOT your group's real answer:
LEGIT_TIME_MIN: 3
```

ATTACK_RESULT

The code the Attacker team reported — should match LEGIT_RESULT if they cracked it correctly.

```
Example only — NOT your group's real answer:
ATTACK_RESULT: 7
```

ATTACK_TIME_MIN

The Attacker team's TOTAL time, from first seeing the ciphertext to reporting their result — this includes their factoring step.

```
Example only — NOT your group's real answer:
ATTACK_TIME_MIN: 9
```

ATTACK_METHOD_NOTE

1-2 lines on HOW the Attacker team actually cracked it — this is your evidence for the two score fields below, not just the final number.

```
Example only — NOT your group's real answer:
ATTACK_METHOD_NOTE: Factored n by trial division (found factors on the 3rd divisor tried),
solved for d the same way the Key Generator did, decrypted with square-and-multiply.
```

How To Score The 3 Judgement Rows

These three fields are the only ones where YOU decide the number, using the guidance below — the same guidance a teacher would otherwise apply. Fill these in before you touch the Result Card.

ATTACK_RESULT_SCORE (0-5)

Full 5 if the Attacker independently reached the SAME correct answer as the Receiver, without ever being told the private key. Partial 2-3 if their method was right but they ran out of time or made an arithmetic slip. 0 if the method itself was wrong.

ATTACK_EFFICIENCY_SCORE (0-3)

Reward a systematic method over guessing — e.g. trying divisors in ascending order (2, 3, 5, 7, 11...) rather than random. Give 3 for a clearly systematic approach, 1-2 if they got there haphazardly or skipped a candidate they should have checked, 0 for no real method.

EXPLANATION_SCORE (0-3)

Before writing this score, pick ONE member at random from each sub-team — not the strongest student — and ask them to explain one step out loud. Score 3 only if they can actually explain it (not just read out a number), 1-2 for a partial explanation, 0 if they can't explain it at all.

Do this before you write the number: The explanation check has to happen live, right now, with a student you pick — not the team's own choice of who answers.

Submitting

How to submit: Hand the completed Result Card page (or a clear photo of it) to your teacher. If asked to submit digitally, type the exact values into a plain text file named Group1_Result.txt, one "FIELD: value" line per field, exactly as labelled above.

n randomly.
e checked, 0

RSA KEY GENERATION

Varun Vinod
30

Role → Key generator

$p = 5 \quad q = 11$

$n = p \times q$

$n = 5 \times 11$

$n = 55$

$\phi(n) = (p-1)(q-1)$

$= 4 \times 10$

$= 40$

$\gcd(3, 40) = 1$

$e = 3$

$(e, n) = (3, 55)$ public key.

$(3 \times d) \bmod 40 = 1$

$d = 27$

VERIFICATION → $3 \times 27 = 81$

$81 \bmod 40 = 1$

$(d, n) = (27, 55)$ Private key.

HANDOFFS

Public key (3,55) → Sender A, B

PRIVATE KEY (27, 55) → Reciever A, B

Dion Boby
Roll No - 10
Role : Sender A

MESSAGE
M = 4
PUBLIC KEY
e = 3

n = 55

ENCRYPTION

$C = M^e \bmod n$

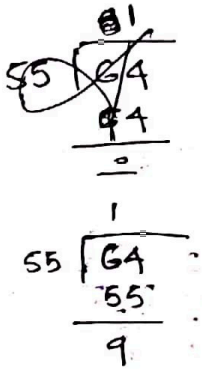
$C = 4^3 \bmod 55$

$4^3 = 4 \times 4 \times 4 = 64$

$C = 64 \bmod 55$

$C = 9$

... FINAL CIPHER



HANDOFF

SENDER A → CIPHER TEXT C=9 → COURIER

~~Dion Bobby~~

Varun P Das

Roll no : 29

Role : Sender B

M = 4 MESSAGE

e = 3 PUBLIC KEY ~~DION~~

n = 55

ENCRYPTION

$C = M^e \bmod n$

$C = 4^3 \bmod 55$

$64 \bmod 55$

$C = 9$ ... Final Cipher

Sender A : C=9

Sender B : C=9 Both Matches.

HANDOFF

SENDER B → Ciphertext C=9 → COURIER

BLIC INFORMATION

= 3

$(e, n) = (3, 55)$ public key

$n = 55$

$C = 9$ Ciphertext From Sender

ANUSREE ANILKUMAR
Roll No: 6
ROLE - COURIER / CHANNEL

Carry $C = 9$ from Sender to Reciver

$C = 9$ travels through a public channel

Attacker intercept $C = 9$.

Rosh K Lyo
Roll No - 26
Role: Receiver A

$p = 5 \quad q = 11$

$n = 55$ RECIEVED CIPHER

$\phi(n) = 40$ $C = 9$

$e = 3$ FROM COURIER.

$(3 \times d) \bmod 40 = 1$

$d = 27$ RECIEVED FROM KEY GENRATOR

$M = C^d \bmod n$

Square and multiply method

$= 9^{27} \bmod 55$

$= 9^1 \bmod 55 = 9$

$= 9^2 \bmod 55 = 26$

$= 9^4 \bmod 55 = 16$

$= 9^8 \bmod 55 = 36$

$= 9^{16} \bmod 55 = 31$

$9^{27} \bmod 55 = 9^1 \times 9^2 \times 9^8 \times 9^{16} \bmod 55$

$= 9 \times 26 \times 36 \times 31 \bmod 55$

$M = 4$

Decrypted Message = 4

HAND OFF

RECIEVER A → Message M=4 Time = 18min → JUDGE / TIMEKEEPER

~~Rosh & Lijo~~

Bibin Alex Tom

Roll no 8

Role : Reciever B

s q = 1 RECIEVED CIPHER C = 9 FROM COURIER

n = 55

$\phi(n) = 40$

e = 3

$(3 \times d) \bmod 40 = 1$

d = 27 RECIEVED FROM KEY GENERATOR.

$M = c^9 \bmod n$

Square and multiply method.

$= 9^{27} \bmod 55$

$= 9^1 \bmod 55 = 9$

$= 9^2 \bmod 55 = 26$

$= 9^4 \bmod 55 = 16$

$= 9^8 \bmod 55 = 36$

$= 9^{16} \bmod 55 = 31$

$9^{27} \bmod 55 = 9^1 \times 9^2 \times 9^8 \times 9^{16} \bmod 55$

$9 \times 26 \times 36 \times 31 \bmod 55$

$(9 \times 26 \bmod 55)\ (36 \times 31 \bmod 55)$

Decrypted message : M = 4

Reciever A : M : 4

Reciever B : M : 4 both Matches.

HANDOFF

RECIEVER B → M = 4 time : 18 min → JUDGE / TIMEKEEPER

Role : Attacker A Adarsh Kumar
30sec 55, 07, 2

Information Intercepted

Ciphertext

$C = 9$

Public key $(e, n) = (3, 55)$

$d = ?$

Factor n, $n = 55$

Try prime no.

$55 \div 2 \rightarrow$ not divisible

$55 \div 3 \rightarrow$ not divisible

$55 \div 5 \rightarrow 11$

$\therefore p = 5 \quad q = 11$

Handoff

Attacker A $\rightarrow p = 5, q = 11 \leftrightarrow$ Attacker B

Aalbin Joseph
Attacker - B

$n = 55$

$\varphi(n) = 40$

$e = 3$

* finding d

the value of d must satisfy $(e \times d) \bmod \varphi(n) = 1$

$$e \times d \equiv 1 \bmod \varphi(n)$$

$e = 3$

$\varphi(n) = 40$

$$3d \equiv 1 \bmod 40$$

this means

$$3d = 40k + 1$$

* trying different value of k

when $k = 1$

$$3d = (40 \times 1) + 1$$

$$3d = 41$$

$$d = 41/3 = 13.67 \quad \times$$

Not an integer.

when $k = 2$

$$3d = (40 \times 2) + 1$$

$$3d = 81$$

$$d = 81/3$$

$$d = 27 \quad \checkmark$$

* Verify the value of d

$e \times d \bmod 40 = 1$

Substitute:

$(3 \times 27) \bmod 40$

$81 \bmod 40$

$81 = 40 \times 2 + 1$

the remainder is: 1

So, $d = 27$

Public key is (e, n)

$(3, 55)$

Private key is (d, n)

$(27, 55)$

HANDOFF

Attackee B → Private key $d = 27$ → Attacker C for decryption.

Joshua Geo Johnson
18
Student 9
Attacker

$C = 9 \quad d = 27 \quad n = 55$

Decrypting

$m = c^d \bmod n$

$= 9^{27} \bmod 55$

SQUARE AND MULTIPLYING

1; 2, 4, 8, 16, 32, 64 .....

$27 - 16 = 11$
$11 - 8 = 3$
$3 - 2 = 1$
$1 - 1 = 0$

} Try to fill with the squares of each no.

$27 = 16 + 8 + 2 + 1$

$= 9^{16} \times 9^8 \times 9^2 \times 9^1 = 9^{27}$

$9^1 = 9 \quad (9 \bmod 55)$

$9^2 = 26 \quad (81 \bmod 55)$

$9^4 = 16$

$9^8 = 36$

$9^{16} = 31$

Substituting value 9^7 in $9^{16} \times 9^8 \times 9^2 \times 9^1 \bmod 55$ --

$= 31 \times 36 \times 26 \times 9 \bmod 55$ --

$= 261144 \bmod 55$

$= 4$

Decrypted message = 4

HANDOFF

Attacker C → Decrypted message M = 4, total time = 18min 30sec

→ Judge / Timekeeper